

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe Wagner earthing device with suitable diagram.

Q.24 Explain block diagram and working principle of a basic CRT and CRO. Also discuss their merits and demerits.

Q.25 Write a short note on following

- a) De Sauty's bridge.
- b) CRO

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Roll No.

**3rd Sem / Instrumentation and Control /
Medical Electronics**

Subject : Measurement and Instrumentation

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which method is used for high resistance measurement.

- a) Megger
- b) Potentiometer method
- c) Both (a) and (b)
- d) None of these

Q.2 In electrodynamic wattmeter moving coils is also called_____.

- a) Current Coil b) Pressure Coil
- c) Both a & b d) None of these

- Q.3 _____ is used to measure temperature.
- a) ammeter b) voltmeter
c) balance d) thermocouple

- Q.4 Unit of power is _____.
- a) Henry b) Weber
c) Watt d) Joule

- Q.5 Unit of Inductance is _____
- a) Henry b) Weber
c) Watt d) Joule

- Q.6 Which part is called the heart of CRO.
- a) CRT b) Sweep Generator
c) Trigger Circuit d) Amplifier

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 _____ is used for medium value resistance m's
- Q.8 Digital frequency meter measures _____.
- Q.9 Expand DSO
- Q.10 Rectifier type meter uses _____ with PMMC.

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- Q.11 Wattmeters Power measurement method uses 2 wattmeter. (True/False)

- Q.12 Expand CRT _____

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Describe m's of resistance.
- Q.14 Describe Maxwell bridge.
- Q.15 Explain principle of Multimedia.
- Q.16 Explain construction and working of Stroboscopes.
- Q.17 Explain permanent Magnet Moving Coil Meters.
- Q.18 Explain Potentiometer method for measurement of resistance.
- Q.19 Explain Block diagram and working principle of a basic DSO. Discuss advantages, disadvantages and applications.
- Q.20 Comparison between three-phase and single-phase system.
- Q.21 Explain Hay's bridge. Discuss their advantages and disadvantages.
- Q.22 Explain thermocouple type meters.

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